ML_Lab7_UEN

October 5, 2026

[1]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.tree import DecisionTreeClassifier
```

[2]:
```
# loading the mnist dataset into the notebook
# each sample in dataset will have 784 features
mnist = fetch_openml('mnist_784')
```

[3]:
```
X_train, X_test, y_train, y_test = train_test_split(mnist.data, mnist.target,
                                                    test_size=0.2,
                                                    random_state=42)
```

[6]:
```
X_train.shape
```

[6]:
```
(56000, 784)
```

[7]:
```
from sklearn import tree
model = tree.DecisionTreeClassifier(criterion="entropy")
model.fit(X_train, y_train)
```

[7]:
```
DecisionTreeClassifier(criterion='entropy')
```

[8]:
```
y_pred = model.predict(X_test)
```

[9]:
```
from sklearn.metrics import classification_report, confusion_matrix,␣
 ↪accuracy_score

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
cr = classification_report(y_test, y_pred)
print("\nClassification Report:",)
print (cr)
acc = accuracy_score(y_test,y_pred)
print("\nAccuracy:",acc)
```

```
Confusion Matrix:
[[1246    0   17    7    9   12   23    5   13   11]
 [   3 1540    7   12    3   10    2   11    9    3]
 [   7   16 1195   36   14   12   22   31   34   13]
 [   5    5   51 1213    9   51    6   26   43   24]
 [   3    3   18    6 1132   10   13   16   22   72]
 [  15   15   18   69   10 1045   26    5   42   28]
 [  15    7   13    9   24   24 1266    5   24    9]
 [   1   10   48   14   16    5    0 1366    6   37]
 [   6   21   30   46   35   31   22   12 1116   38]
 [   9    4   13   25   76   26    9   24   27 1207]]

Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.93      0.94      1343
           1       0.95      0.96      0.96      1600
           2       0.85      0.87      0.86      1380
           3       0.84      0.85      0.85      1433
           4       0.85      0.87      0.86      1295
           5       0.85      0.82      0.84      1273
           6       0.91      0.91      0.91      1396
           7       0.91      0.91      0.91      1503
           8       0.84      0.82      0.83      1357
           9       0.84      0.85      0.84      1420

    accuracy                           0.88     14000
   macro avg       0.88      0.88      0.88     14000
weighted avg       0.88      0.88      0.88     14000


Accuracy: 0.8804285714285714
```

[10]:
```
model = RandomForestClassifier(bootstrap=True, criterion='gini',
            min_samples_leaf=1,
            min_samples_split=2, n_estimators=20)

model.fit(X_train, y_train)
```

[10]:
```
RandomForestClassifier(n_estimators=20)
```

[11]:
```
y_pred = model.predict(X_test)
```

[12]:
```
from sklearn.metrics import classification_report, confusion_matrix,␣
 ↪accuracy_score

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
```

```
print(cm)
cr = classification_report(y_test, y_pred)
print("\nClassification Report:",)
print (cr)
acc = accuracy_score(y_test,y_pred)
print("\nAccuracy:",acc)
```

```
Confusion Matrix:
[[1326    0    1    0    1    2    6    1    5    1]
 [   0 1578    6    3    2    2    1    5    2    1]
 [   6    4 1326    6    7    2   10   10    9    0]
 [   2    1   28 1337    3   23    0   15   18    6]
 [   5    1    2    1 1247    0    4    2    4   29]
 [   6    1    3   29    3 1204    9    1   12    5]
 [   7    1    2    0   10   10 1364    0    2    0]
 [   4    5   18    1    8    0    0 1442    6   19]
 [   2    5   14   18    7   22    6    7 1267    9]
 [   6    5    5    9   21    6    2    8   17 1341]]

Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.99      0.98      1343
           1       0.99      0.99      0.99      1600
           2       0.94      0.96      0.95      1380
           3       0.95      0.93      0.94      1433
           4       0.95      0.96      0.96      1295
           5       0.95      0.95      0.95      1273
           6       0.97      0.98      0.97      1396
           7       0.97      0.96      0.96      1503
           8       0.94      0.93      0.94      1357
           9       0.95      0.94      0.95      1420

    accuracy                           0.96     14000
   macro avg       0.96      0.96      0.96     14000
weighted avg       0.96      0.96      0.96     14000


Accuracy: 0.9594285714285714
```

Exercise: Build Random Forest Classifer to predict class label of EMNIST dataset

0.1 Lab activity:

Task1 The EMNIST dataset is a set of handwritten character digits derived from the NIST Special Database 19 and converted to a 28x28 pixel image format and dataset structure that directly matches the MNIST dataset. Further information on the dataset contents and conversion process can be found in the paper available at https://arxiv.org/abs/1702.05373v1.

Generate classification report, confusion matrix and accuracy on the test set (you may consider 30% of the data for testing) for both Decision tree and Random forest classifier.

Task 2 — Investigate Bootstrap Sampling Objective

Understand why Random Forest uses bootstrap samples and how different training sets lead to different trees. Use bootstraps of size 5k (any random samples) without any random feature selection.

Train a random forest with

n_estimators = 5

n_estimators = 20

n_estimators = 50

Task 3: Random Feature Selection Do the above experiments using random feature selection (fix the training samples common across all bootstraps)

For task 2 and 3, use breast_cancer data which we have used in the previous lab activity.

[ ]: